

Program -9-Documentation

Title: Organize Kubernetes pods using Labels and Namespaces

Objective: Create isolated environments using namespaces and manage pods efficiently using labels

1. Project Structure

```
Prgm9/  
|  
|--- namespaces.yaml  
|--- pods.yaml
```

2. Configuration files

Create **namespaces.yaml**

```
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

Create **pods.yaml**

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: frontend-pod  
  namespace: dev  
labels:  
  app: frontend  
  env: dev
```

```
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
```

```
    ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

3. Apply Namespace YAML and Pods YAML

Creates dev and prod namespaces

```
kubectl apply -f namespaces.yaml
```

Pods created inside respective namespaces

```
kubectl apply -f pods.yaml
```

4. Verify Pods by Namespace

```
kubectl get pods -n dev
```

```
kubectl get pods -n prod
```

5. Filter Pods by Label

Shows how labels help filtering

List frontend pods in dev

```
kubectl get pods -n dev -l app=frontend
```

List backend pods in prod

```
kubectl get pods -n prod -l app=backend
```

```
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get namespaces
NAME                STATUS    AGE
default             Active   8m11s
kube-node-lease     Active   8m9s
kube-public         Active   8m11s
kube-system         Active   8m11s
kubernetes-dashboard Active   7m22s
prod                Active   6m43s
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl apply -f pods.yaml
pod/frontend-pod unchanged
pod/backend-pod unchanged
Error from server (NotFound): error when creating "pods.yaml": namespaces "dev" not found
Error from server (NotFound): error when creating "pods.yaml": namespaces "dev" not found
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ nano pods.yaml
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ nano namespaces.yaml
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ nano pods.yaml
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod unchanged
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod unchanged
pod/backend-pod unchanged
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          7s
frontend-pod  0/1     ContainerCreating   0          7s
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          11m
frontend-pod  1/1     Running   0          4m47s
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-brffc  1/1     Running   0          9m10s
nginx-deployment-6f9664446b-vwp28  1/1     Running   0          9m10s
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          2m54s
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          14m
1rv24nc082_rajat@rajat@vostro-5581:~/devops/labprogram/prgm9$
```